

PRINCE CHAUHAN

SENIOR QUANTITATIVE ANALYST | SYSTEMATIC DERIVATIVES & STATISTICAL ARBITRAGE

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SEBI Registered Research Analyst (Reg. No. INH100006843) | NISM Series-XV Certified

Public Quantitative Suite: github.com/pchauhanrajputpc-sketch/quant-research-portfolio (29/29 Passing CI Tests)

PROFESSIONAL SUMMARY

Senior Quantitative Analyst and Systematic Derivatives Strategist with **9+ years of experience** developing, validating, and deploying options volatility arbitrage, statistical cointegration, and microstructure-aware execution architectures across Indian (NSE/BSE) index and equity derivatives. Specializes in implied volatility surface modeling (SVI & cubic splines), Johansen multivariate cointegration, unsupervised regime classification, and causal point-in-time backtesting engines with explicit turnover friction (0.50%), regulatory margin calibration (INR 2.5L / 1.8L / 1.0L), walk-forward optimization, and 1,000-path Monte Carlo stress testing. Combines quantitative programming rigor in Python (Polars, NumPy, SciPy) with advanced financial accounting forensics (CA Final G1, B.Com Hons 1st Div).

PROFESSIONAL EXPERIENCE

SMC Global Securities Ltd. - Senior Quantitative Analyst, Systematic Strategies & Quant Desk Dec 2024 - Present

- Architect causal 1-minute event-driven execution loops and research data pipelines for systematic intraday and multi-day options strategies across Nifty, Bank Nifty, Sensex, and single-stock derivatives with zero lookahead bias.
- Formulate Bank Nifty basket statistical arbitrage models applying Johansen cointegration rank tests, dynamic rolling beta hedging, and Ornstein-Uhlenbeck spread half-life modeling across heavyweight banking equities.
- Calibrate systematic options models achieving 2.5+ Out-of-Sample Sharpe and 2.1+ Profit Factor in chronological simulations deducting explicit 0.50% turnover friction (brokerage, STT, exchange fees, SEBI turnover, GST, and 1-tick slippage).
- Engineer asynchronous market-data caching and order routing using Python Asyncio, broker WebSockets, and memory-mapped (mmap) IPC, capturing 1.8 bps in market-crossing savings and eliminating API rate-limit bottlenecks.
- Structure real-time portfolio Greeks (Delta, Gamma, Vega, Theta), dealer Gamma Exposure (GEX) mapping, parametric/historical VaR, automated volatility kill switches, and SPAN cross-margin risk models.

Smart Bulls Capital - Research Analyst, Derivatives & Quantitative Research Mar 2017 - Nov 2024

- Researched, structured, and traded systematic multi-leg options volatility strategies (calendar spreads, ratio butterflies, iron condors) with dynamic delta and vega rebalancing driven by real-time IV surface spline smoothing.
- Formulated Black-Scholes-Merton PDE inversion models (Newton-Raphson/Brent), cubic spline & SVI strike smoothing, and real-time dealer Gamma Exposure (GEX) profiles to map institutional pin risk.
- Conducted financial statement forensics, discounted cash flow (DCF) modeling, and equity valuation across banking, energy, and industrial sectors to support desk capital allocation.
- Supervised daily P&L attribution, margin utilization, order slice execution, and drawdown limits across multi-strike options books.

SELECTED QUANTITATIVE RESEARCH & SYSTEMS

- Point-in-Time Backtesting Engine & 11-Stage Validator Gate [CI: 9/9 Tests Pass]:** Built modular Python framework utilizing 7+ years of causal 1-minute historical data with zero lookahead bias; pre-deducts 0.50% turnover friction, enforces regulatory margin models (INR 2.5L/1.8L/1.0L), 11-stage audit gate (Rules L1-L22), 70/20/10 Walk-Forward Optimization, and 1,000-path Monte Carlo drawdown cones.
- Arbitrage-Free Options Volatility Surface & GEX Profiling [CI: 5/5 Tests Pass]:** Formulated Black-Scholes inversion using Newton-Raphson/Brent solvers, cubic spline & SVI surface fitting, arbitrage-free total variance constraints ($dW/dK \geq 0$), and dealer Gamma Exposure (GEX) sensitivity profiling.
- Bank Nifty Basket Cointegration & Statistical Arbitrage [CI: 4/4 Tests Pass]:** Implemented Johansen multivariate cointegration test, dynamic rolling OLS hedge ratios, Ornstein-Uhlenbeck spread half-life estimation, and mean-reverting Z-score execution across major banking constituents.
- Market Regime Volatility Gating & Microstructure Simulator [CI: 7/7 Tests Pass]:** Engineered unsupervised K-Means clustering on Parkinson RV, IV/RV ratios, and return skewness, reducing simulated maximum drawdown by 28%; paired with an execution simulator modeling order book queue priority and square-root market impact.

TECHNICAL & QUANTITATIVE COMPETENCIES

- Quantitative Research:** Statistical Arbitrage, Johansen Cointegration, Dynamic Beta Hedging, SVI Volatility Surface Fitting, Implied & Realized Volatility (IV/RV, Parkinson), Market Regime Detection (K-Means), Walk-Forward Optimization (WFO), Monte Carlo Simulation.
- Derivatives & Risk Governance:** Black-Scholes Pricing, Portfolio Greeks (Delta, Gamma, Vega, Theta), Delta/Vega Neutrality, Dealer Gamma Exposure (GEX), Parametric & Historical VaR, SPAN Cross-Margin Optimization, Exchange Margin Benchmarks.
- Programming & High-Performance Data:** Python 3.11+, Polars, DuckDB, NumPy, SciPy, Statsmodels, Scikit-Learn, VectorBT, SQL, Apache Parquet, Asyncio, WebSockets, Memory-Mapped IPC (mmap), ReportLab PDF Engine, Git, Linux/Bash.
- Markets & Microstructure:** NSE/BSE Equity & Index Derivatives (Nifty, Bank Nifty, Sensex), Stock Options/Futures, Limit Order Book Dynamics, Smart Limit Slicing, Fill Slippage Minimization, CBOE SPX/VIX Benchmarks.

EDUCATION, ACCOUNTING FORENSICS & REGULATORY CREDENTIALS

- Bachelor of Commerce (Honours):** Kirori Mal College, University of Delhi (2011 - 2014) | 71% Marks (First Division)
- Chartered Accountancy (CA Final):** Institute of Chartered Accountants of India (ICAI, 2018) | 78% Marks (Cleared Group 1)
- Regulatory Licenses:** SEBI Registered Research Analyst (#INH100006843) | NISM Series-XV: Research Analyst Certified
- Earlier Experience:** Corporate Finance Executive at Indian Oil Corporation Ltd. (IOCL, 2016 - 2017) | Finance Executive at CA Matta & Associates (2014 - 2016)